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SEAT No. :

P1

[Total No. of Pages : 2

**APR-18/T.E./Insem - 1**  
**T.E. (Civil) (Semester - II)**  
**ADVANCE SURVEYING**  
**(2012 Pattern)**

*Time : 1 Hour]*

*[Max. Marks : 30*

*Instructions to the candidates :*

- 1) *Neat diagrams must be drawn wherever necessary.*
- 2) *Figures to the right indicate full marks.*
- 3) *Use of logarithmic tables slide rule, mollier charts, electronic pocket calculator and steam tables is allowed.*
- 4) *Assume suitable data, if necessary.*

**Q1) a)** Explain with neat sketches, commonly used layouts of triangulation systems. **[5]**

b) State any four advantages of space based positioning systems. **[5]**

OR

**Q2) a)** What factors are to be considered while selecting a best triangulation figure or system? **[4]**

b) Two triangulation stations A and B are 42 km apart and have elevation of 279 and 276 m respectively. Find the minimum height of signal required at B so that the line of sight may not pass nearer the ground than 3 m. The intervening ground may be assumed to have a uniform elevation of 252 m. **[6]**

**Q3) a)** Explain the term sounding and explain any two methods of locating the sounding positions. **[4]**

b) The observations were made on three stations A, B, C from boat at O with help of sextant. Stations B & O being on the same side of AC. Calculate the distances of the boat from three stations. Angle AOB =  $30^{\circ}25'$ , Angle BOC =  $45^{\circ}25'$ , Angle ABC =  $130^{\circ}10'$ , AB = 4000 m, BC = 4995 m. **[6]**

OR

**P.T.O.**

- Q4)** a) What do you understand by three point problem? Explain with neat sketch. [5]  
b) What do you mean by echo- soundings, explain with sketch. [5]

- Q5)** a) The triangulation stations A & B are 2800 m apart. Observations were for vertical angle of elevation from A to B and the mean angle observed was  $1^{\circ} 28' 32''$ . The height of the instruments was 1.38 m and signal was 2.46 m high. If the reduced level of station A was 125 m and the coefficient of refraction was 0.07, calculate the reduced level of B. The radius of the earth 6372 km. [6]  
b) While doing an underground survey describe the transferring the surface alignment through a Shaft? [4]

OR

- Q6)** a) Derive the expression for the difference of level between two points A and B a distance D apart, with the vertical angle as the angle of elevation from A to B. [6]  
b) Explain with neat sketch, how determine the location of piers of bridge. [4]

